MINING DEALER BEST PRACTICE

Machine Shop Components Lifting Operations Assurance Strategy

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The process of component lifting at the machine shops differs from the equipment disassembly and assembly process (as illustrated on SISWEB) due to the final mounting location that the machining equipment has (see Figure 1), this fact forces the technician to perform potentially risky movements, involving also the machine and even the component itself.

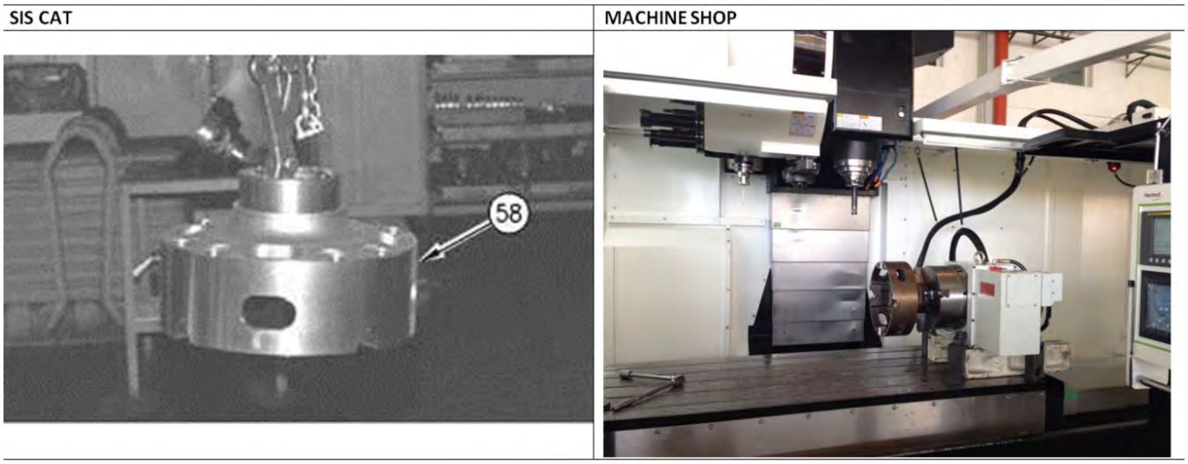


Figure 1. SISWEB Horizontal Lifting (assembly and disassembly) versus the vertical installation in the machining tool

Besides, earlier in the component lifting process, the correct selection of accessories, chains or slings depended on the technician´s knowledge and their availability to be on-site; so they did not always perform the lifting procedure properly. This put the physical integrity of the components, machines, and also the technician in danger.

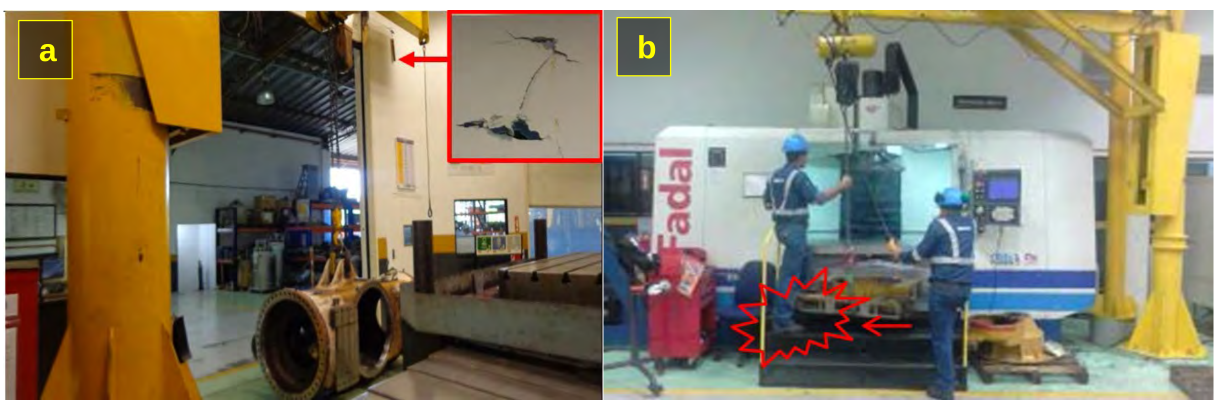


Figure 2. Frequent accidents due to inadequate lifting movements. a) Damage to the property; b) blow to the technical staff.

This Best Practices aims to improve the steps that must be taken into account when performing a safe lifting procedure, and transporting components inside the machine shop. This also considers the final installation positions on the different machining devices and the increase of safety in these tasks.

2.0 Best Practice Description

The improvement is to establish a methodical strategy for lifting and transporting component, taking into account the final position of the component installation on different machines (Mounting mill boring machine, lathe, etc.). By defining and standardizing the lifting elements that are required in each of the repair processes, this aims to reduce the potential risk and accident that can cause casualties or damage to the property and to speed up the process of moving components.

This strategy consists of:

- Illustrative posters in work areas
  - They are located on each machine, showing the most common working components of these, identifying their P/N and weight.

- Lifting manual
  - It's guidebook for the component lifting safely and simply.
  - This manual consists of the following parts:
    - a) P/N components index of higher turnover in the shop
    - b) List of lifting elements capacity (shackles, eyebolts, chains, etc.)
    - c) Components weights Listings (SISWEB)
    - d) Lifting calculations reports for each component (under international standards)

- Components lifting Kit shelves.
  - The shelves are located closest to each of the machines; there you can find all the lifting elements organized by kits, according to the lifting manual. See Fig. 3

Figure 3. - 1. Illustrative posters; 2. Lifting Manual; 3. Lifting kit shelves

3.0 Implementation Steps

Planning:

- ✓ We had access to all the information regarding parts or components with a higher turnover.
- ✓ A statistical analysis was conducted and finally a list of parts by P/N was acquired for each of the machining devices at the shop.

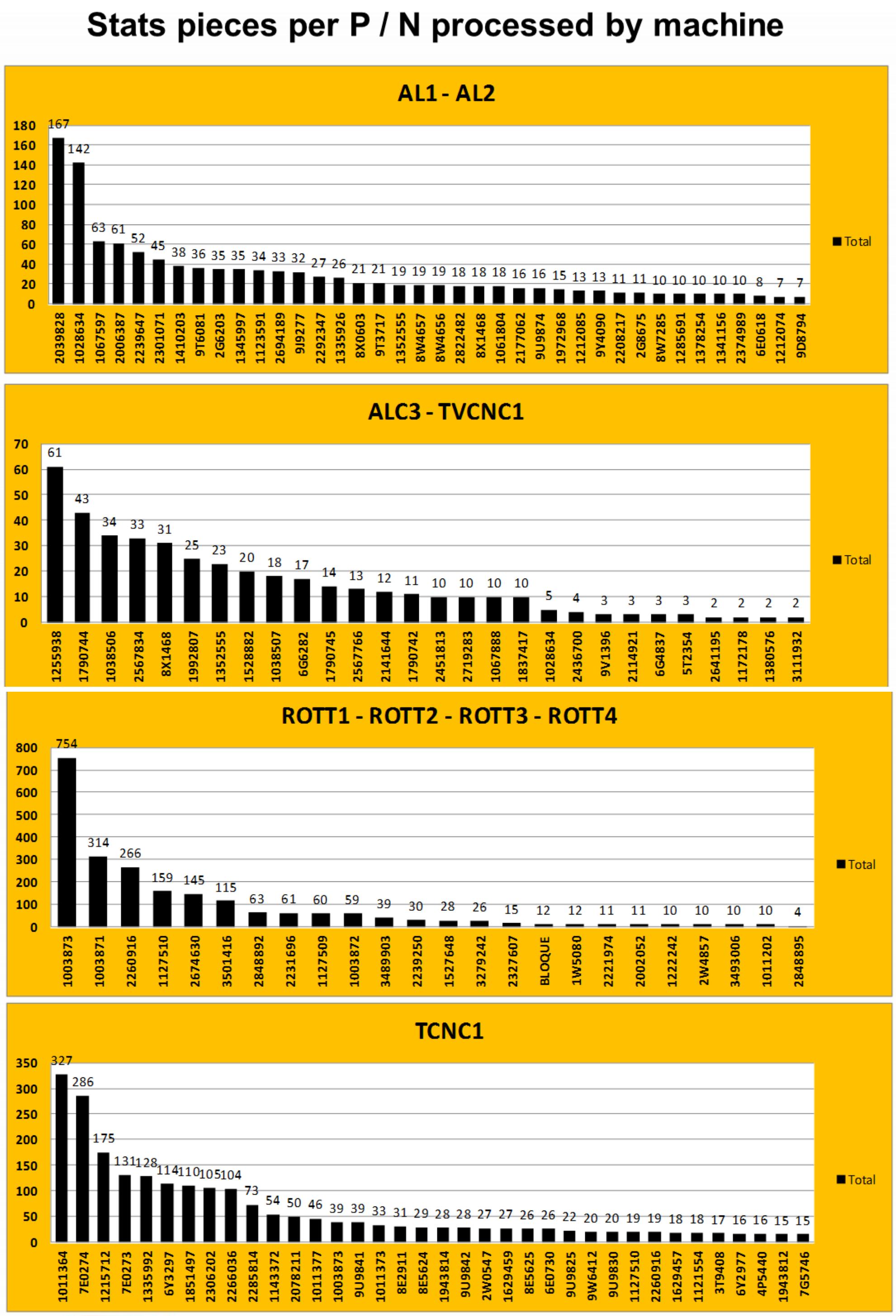


Figure 4. Example of the statistical analysis of components of higher turnover in the shop

- ✓ The weight and the lifting recommendations (for assembly and disassembly processes) of each component were sought thru SISWEB identifying the P/N of the tooling and lifting accessories used as well.

SIS:

-NPR
-ASSEMBLY DISASSEMBLY MANUALS (CONTENT).

Access Methods - Product ID Not Required

- Advanced Full Text Search
- Media Search
- Similar Parts Search
- NPR
- Parts List
- Kits Information
- Engine Performance Specifications
- Service Software Files
- Other Repair Process Information
- Service Forms

EX (NPR):
PN: 8X0603 CARRIER PLANETARY
WEIGHT: 580 KG

Part Number: 8X0603 Country: none Eng Change: Search

For a complete list of service information for this part, please refer to Product Structure in SIS.
Numerical Parts Record Inquiry

8X0603 00

Part Number: 8X-0603 Change: 00 Canceled NPRs: 1 Weight:1118Lbs., 580.01 Kg

EX ASSEMBLY DISSASEMBLY MANUAL CONTENT:
PN: 8X0603 CARRIER PLANETARY
WEIGHT: 580 KG

Information Types

Contamination Control Guidelines
Disassembly and Assembly
Information Release Memo
Operation and Maintenance Manual
Parts Identification
Reuse And Salvage Guidelines
Service Magazine

Results - Auto Filter

- Final Drive, Brake and Wheel (Rear) - Disassemble 01.06.2004
  - Disassembly Procedure 01.06.2004
- Final Drive, Brake and Wheel (Rear) - Assemble 01.06.2004
  - Assembly Procedure 01.06.2004

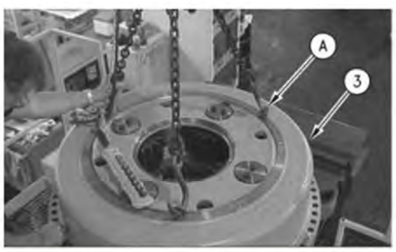


Illustration 3 g00815456

3. Use Tooling (A) and a suitable hoist to remove planetary carrier (3). The weight of planetary carrier (3) is 580 kg (1279 lb).

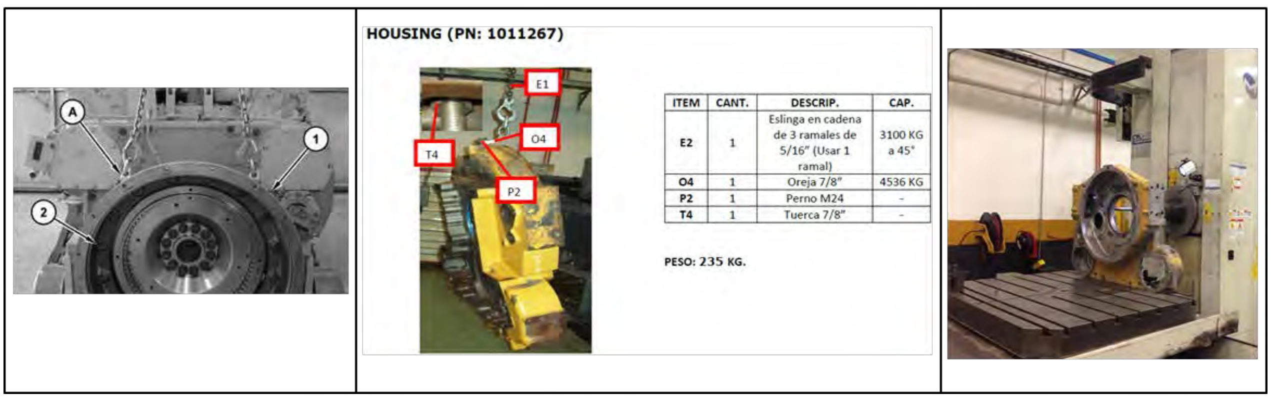
PESOS DE SUBCOMPONENTES

PIEZA	NÚMERO DE PARTE	PESO (KG)
Anchor Brake	1038507	200
Arm Center	1355269	168
Axle	1172178	371
	1172158	215
Bar	8W7285	81
	1172152	125
Bastidor	1380576	2041
	1123036	982
	1123038	898
	2123499	2223
Bogie	1573107	1683
	2618288	1465
	1384151	590
	9W7871	533
	6T1860	444
	9W0683	668
	1342680	726
Bracket	9Y9388	47
	1W0533	6
	3258123	47
	6N7328	22
Cage	2379848	32
	1283880	42
Carrier	8X0603	508
	1T1819	57
	5T9058	420
Case	9U9874	276
Cover	7G6520	10
	2000818	20
	5V4703	55
Cylinder	2039828	237
	2239647	288
	9J9277	288
	9T6081	196
	2292347	243
Drawbar	6G4837	938
Elbow	6N4247	16
	7C2023	21
Head	6E0730	170
	4T4838	165
	2239652	60
	1026340	62

Figure 5. Data on the SIS Web and final listing of main components with P/N and weights.

- ✓ Some training was given in areas such as: Safe handling and lifting of overhead cranes under international standards to one engineer and one supervisor for every 30 components listed above.
- ✓ This trained personnel conducted field work. The safe lifting mode of each component in each machine tool was checked, considering the position used in these in accordance with the most frequent repairs.
- ✓ The memory calculation was performed for each specific lifting operation, under the standards:
- ✓ ASME B30.9 slings
- ✓ ASME B30.26 Rigging Hardware
- ✓ or OSHA 1910.184 Slings
- ✓ The tooling to be used was identified and classified in each areas and the cabinets to store the kits were made.

SIS CAT	HANDBOOK	MOUNTING MILL BORING MACHINE



ITEM	CANT.	DESCRIP.	CAP.
E2	1	Eslinga en cadena de 3 ramales de 5/16" (Usar 1 ramal)	3100 KG a 45°
O4	1	Oreja 7/8"	4536 KG
P2	1	Perno M24	-
T4	1	Tuerca 7/8"	-

Figure 6. SISWEB Research work; field work and lifting elements identification and calculations; Lifting movements assessment for the installation on the machine.

Implementation

- ✓ A lifting manual or a guide showing the list of components is issued. Such listing includes the components weight and also the kits for each machine specific components at the shop. A manual or guide is printed for each area (divided into machines groups).
- ✓ Some acrylic boards are located in each area (Component name, P/N and weight) per machine.
- ✓ The technicians were trained on safe lifting conditions and lifting manual handling as a result of this strategy.
- ✓ The route of the process is as follows:

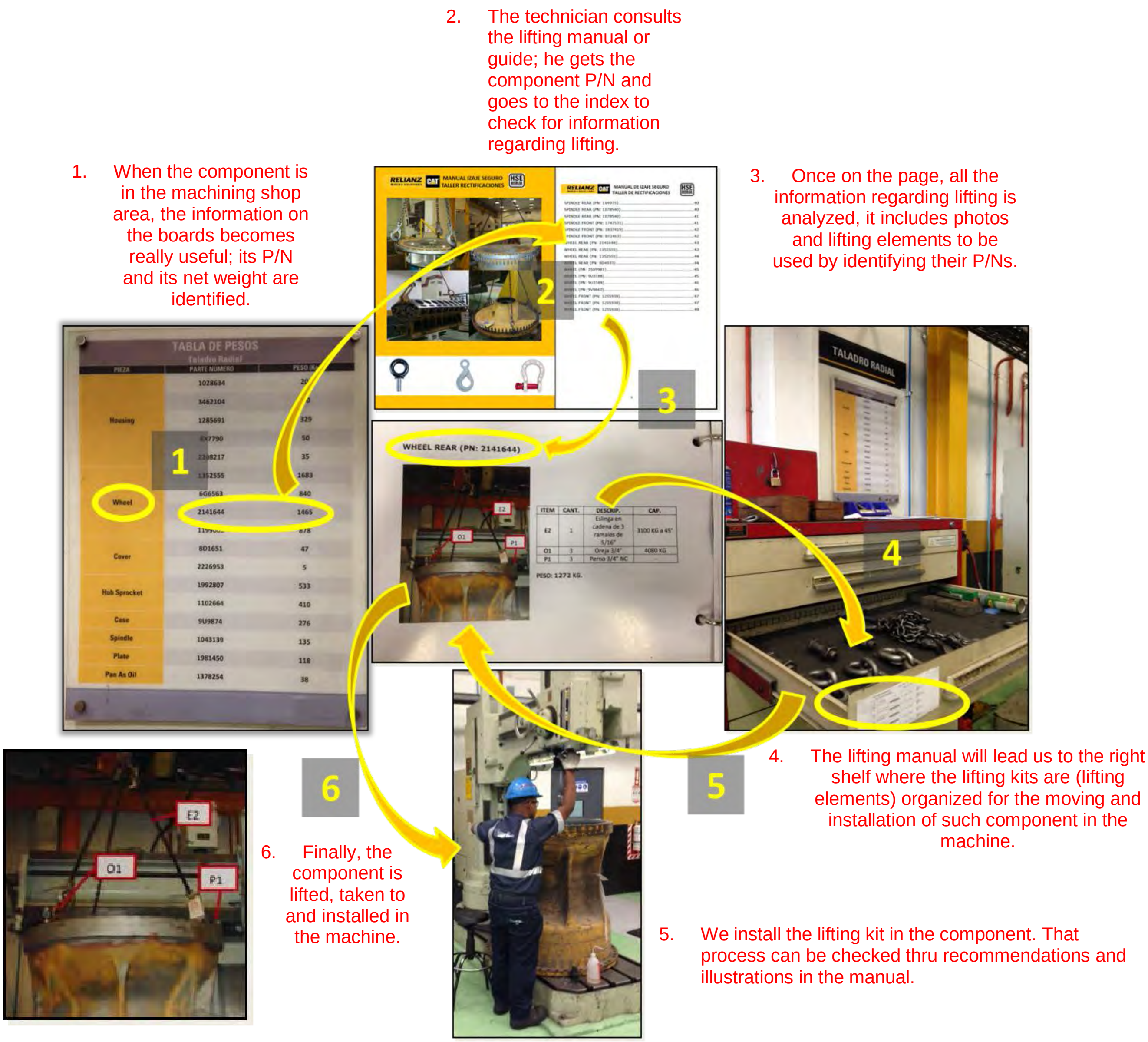


Figure 7. Execution Process of the lifting operations assurance strategy in the machine shop

4.0 Benefits

- 32.2% time reduction in 6 months regarding canvas slings and lifting accessories consumption; it went from $ 7,023.56 USD to $ 4,761.00 USD, considering that the second cost includes the sum of canvas sling consumption after the upgrade ($ 2.218,66USD), plus the initial investment made by switching to chain slings ($ 2.542,34USD, longer service life) see figure 8.

	CANVAS SLING COST		TOTAL COST
2014	MARCH	$40,36	$7.023,56
	APRIL	$739,84	
	MAY	$3.126,19	
	JUNE	$1.367,60	
	JULY	$1.749,57	
	AUGUST	$0,00	
	SEPTEMBER	$1.416,10	$2.218,66
	OCTOBER	$130,34	
	NOVEMBER	$0,00	
	DECEMBER	$70,91	
2015	JANUARY	$601,31	
	FEBRAURY	$0,00	

INITIAL INVESTMENT CHAIN SLING
$ 2.542,34

% Reduction of consumption canvas slings	**68,4%**
cost of investment and consumption chain sling after improvement	**$4.761,00**
% Cost reduction including the initial investment in chain slings	**32,2%**

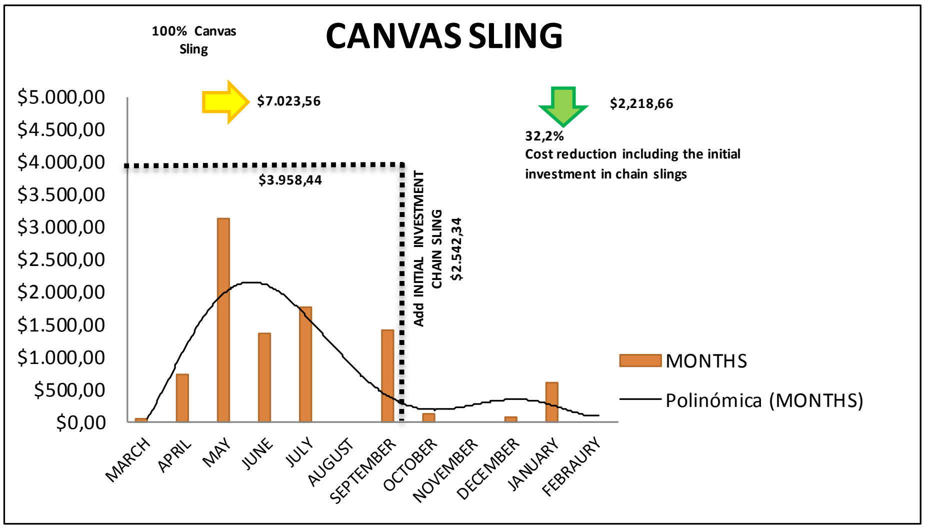


Figure 8. Costs reduction stats per lifting elements consumption with the implementation of the strategy.

- Accident reduction per lifting operations. The six-month term before the upgrade 4 minor accidents were reported; three of them including personal injuries and another one property damage. After the upgrade (Sept. 2014) and up to now; we have not had any accidents, which means a total cost per accident of USD $ 4,712.73. See Figure 9.

BRAND	EQUIPMENT	EQUIPMET COST	COST PER HOUR	REPAIR COST	HOUR DOWN	SICK LEAVE	COST PER ACCIDENT
FD-FADAL	CENTRO DE MECANIZADO VERTICAL CNC 1	$ 75.000,00	$ 6,14		1	1	$ 606,14
MILLTRONICS	TORNO PARALELO CNC 2 MILLTRONICS	$ 264.075,08	$ 9,21				$ -
EASTAR	TORNO CNC #5	$ 350.000,00	$ 16,32				$ -
HANKOOK	TORNO VERTICAL CNC 1 HANKOOK	$ 692.545,10	$ 22,05				$ -
ROTTLER	RECTIFICADORA VERTICAL CNC 1	$ 526.695,12	$ 13,99	$ 410	4,5		$ 472,95
DOOSAN	ALESADORA CNC #1	$ 793.671,85	$ 25,71		1	5	$ 3.025,71
MILTRONICS	ALESADORA CNC #2	$ 392.244,59	$ 16,45				$ -
HULTSFRED	ALESADORA 2	$ 45.000,00	$ 5,95				$ -
WEBSTER BENNET	TORNO VERTICAL 2	$ 100.000,00	$ 7,94		1	1	$ 607,94
MILLTRONICS	FRESADORA 1 CNC	$ 66.289,71	$ 3,27				$ -
							$ 4.712,73

- An approximately 15% time reduction in the lifting operations, mainly due to:
    - No waste of time due to decision making for the selection of lifting elements.
    - There's sorting and organization of lifting kits on site per component and machining device.

5.0 Resources Required

- Statistics of components of higher turnover in the shop
- SIS WEB
- 1 engineer and 1 supervisor for every 30 components.
- Certified Training under lifting processes standard and selection of elements.
- $ 4,800 USD for boards printing, lifting manual and the investment in canvas slings changing (this depends on the capacity of the shop).
- $ 8,300 USD for costs of lifting kits storage shelves (this depends on the work capacity of the shop).

6.0 Supporting Attachments / References

- Lifting manual (for example)

7.0 Related Best Practices

NONE

8.0 Acknowledgements

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